INCH-POUND

MIL-PRF-32009
10 Jul 97

PERFORMANCE SPECIFICATION

CRANE, PORTABLE, FOLDING, 6000 LB CAPACITY,

AIRCRAFT ENGINE AND PROPELLER;

GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers a heavy duty crane that can be used for C-130 engine and propeller removal/installation.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommend for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

SPECIFICATIONS-NONE

STANDARDS-NONE

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted should be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS should be those in effect on the date specified in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/LVF, 225 Ocmulgee Court, Robins AFB, GA 31098-1647, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3950

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J534	Lubrication Fittings
J925	Minimum Service Access Dimensions for Off-Road Machines
J959	Lifting Crane, Wire-Rope Strength Factors

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related detail specifications or specification sheets), the text of this specification should take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample should be subjected to first article inspection in accordance with 4.3 and 6.2.1.

3.2 Materials. Materials not covered by this specification or referenced document should be suitable for the intended environment and application. The contractor shall not use nonstandard or proprietary materials and processes without prior government approval.

3.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used. Surface preparation, finishes and coatings shall be in accordance with (IAW) Contractor's best commercial practices.

3.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with-each other.

3.2.3 Primers and topcoat. Item shall be both primed and top coated with polyurethane.

3.2.4 Drainage. Complete drainage shall be provided for both internal and exterior pockets, areas, parts and components.

3.2.5 Magnesium. Magnesium and magnesium alloys shall be subject to approval by the government.

3.2.6 Prohibited materials. The use of wood and/or wood products is prohibited. Similarly, vinyl, polyvinylchloride, polyester, and corrosive type room temperature vulcanizing (yields acetic acid upon curing) are prohibited for use in all forms, interior and exterior. Use of prohibited materials increases risk of hazardous fumes and fire.

3.2.7 Fasteners. Fastener installation shall be IAW best commercial practices. Fastener quality shall be IAW best commercial standards.

3.2.8 Sealants for permanent fasteners. All joints, rivets and mating/faying surfaces of exterior and interior components shall be sealed using best commercial practices.

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3.2.9 Welding. Ferrous and aluminum welding shall be IAW best commercial practices.

3.2.10 Design and construction. The crane shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations, and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. The design of the crane shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, and replacement. Crane shall be able to meet Air Force requirements for towed and manually propelled support equipment (see 6.3.2). The crane, in any loading condition, including air transport configuration, shall be designed so that it can be maneuvered (pushed, pulled, and rotated) on a level paved surface by no more than two personnel (see 6.3.3); manual forces required to be applied initially to the crane to set it in motion, to be rotated, or to be sustained over short distances (approx. 15 feet), shall not exceed a total of 160 pounds.

3.2.2.1. Reliability. The contractor shall establish, conduct, and maintain a comprehensive reliability program. The crane, as a system, shall have a mean-cycles-between-failure (MCBF) of not less than 1,000 operation cycles demonstrated at a 0.8 confidence level (1,610 operation cycles) in accordance with 4.7. An operation cycle is defined as attaching a load of 6,000 pounds to the cable hook with the crane boom fully extended, lifting the load to maximum hook height, and lowering the load back to ground level.

The reliability demonstration in paragraph 4.6 can be waived only if the contractor can demonstrate that the crane winching system, such as the electrical winch, cable, hook, and etc., has a total combined mean-time-between-failure (MTBF) of not less than 10,000 hours from commercial component test data.

For this specification, MCBF is defined as the mean number of operating cycles during which all parts of the item perform within their specified limits, during a particular measurement interval under stated conditions, i.e., without a failure. A failure is the event, or inoperable state, in which any item or part of an item does not, or would not, perform as specified. The crane shall be designed for a life expectancy of at least 20 years.

3.2.2.2. Maintainability. The crane shall be designed and constructed to provide the following:

a. Servicing, for components requiring scheduled service, without disturbing adjacent components.

b. A minimum number of parts consistent with the performance required herein.

c. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the crane. Where practical, components shall be located for rapid inspection and recognition of potential failure. Service access dimensions shall be in accordance with SAE J925, for servicing while wearing arctic mittens and clothing and chemical defense gear.

d. Ease of adjusting, servicing, and replacing parts and components.

e. Use of standard mechanic's common hand tools found in the Standard Tools for Aeronautical Maintenance 2 (STAM2) list and crane maintenance shop equipment for maintenance.

f. Maintenance with a minimum number of tools selected from the Standard Tools for Aeronautical Maintenance (STAM) list.

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g. Mean-time-to-repair (MTTR) of no greater than 1 hour. MTTR is the sum of corrective maintenance times at any specific level of repair, divided by the total number of failures within an item repaired at that level of repair, during a particular interval under stated conditions. Corrective maintenance is defined as all actions performed, as a result of failure, to restore an item to a specified condition and, as a minimum, includes localization, isolation, troubleshooting, disassembly, interchange, reassembly, alignment, and checkout.

h. Maximum repair time of no greater than 8 hours.

i. Daily operator scheduled maintenance shall be limited to visual inspection(s), and checking and adjusting of fluid levels; inspection points shall not be obstructed or require disassembly or removal to gain access.

j. Designs giving consideration to modular system components and quick disconnect type connectors.

k. The use of special tools is not encouraged. Any special tools must have written approval from procuring agency. The contractor shall provide any special tools, i.e., tools peculiar to this crane, that are not available commercially, needed to perform scheduled or unscheduled maintenance. One set of special tools shall be supplied with each crane in an attached lockable toolbox.

3.2.2.2.1 Maintenance tasks. The crane shall be designed so two mechanics using common hand tools, found in the Standard Tools for Aeronautical Repair (STAR) list, can perform the following maintenance tasks within the time specified:

a. Remove and replace the winch (1 hour).

b. Remove and replace wire rope (1 hour).

c. Remove and replace casters (20 minutes each caster).

d. Daily operator scheduled maintenance and service tasks in 3.2.2.2.i (15 minutes). These tasks shall not require the use of tools.

3.2.2.3 Safety. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.2.2.4 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Safety wire is not permitted.

3.2.2.5 Servicing provisions. Drains, grease fittings, lines, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.2.2.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or

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assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.2.2.7 Component protection. All space in which work is performed during operation, service, and maintenance of the crane shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts, that are of such nature or so located as to be hazardous to personnel, shall be guarded to the extent necessary to eliminate the hazard.

3.2.2.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the crane.

3.2.2.9 Bonding. Metal parts shall be specifically bonded together to minimize the possibility of sparks.

3.2.3 Environmental conditions. The crane shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full time exposure to temperatures ranging from -65^0F to $+125^0$F.

b. Full time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm falling at a rate of 13 cm per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80cm^2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1,750 ft/min, concentration 0.5 g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5,700 ft/min, concentration 0.0773 g/ft^3.

3.2.4 Weight distribution. The weight distribution of the crane in any loading configuration shall not be unbalanced or improperly distributed.

3.2.5 Air transportability. The crane shall be rolled on and off aircraft while the crane is in the folded air transport mode. The crane shall be air transportable without shoring in Types C-130, C-141, C-5, KC-10, KC-135 and C-17 Aircraft (see 6.3.4).

3.2.5.1 Tiedowns. The crane shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in 3.1.5. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.2.6 Factor of safety. Except as otherwise specified, all load bearing components shall be designed with a factor of safety of three, based on the minimum material yield strength.

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3.2.7 Folding and erection. The design of the crane shall allow the crane to be folded into the air transport mode. The time required to prepare the crane, with a maximum of 2 people, into the air transport mode shall not be greater than 30 minutes. The time required to erect the crane, with a maximum of 2 people, into the operating mode shall not be greater than 30 minutes. The use of another crane to assist in folding and erection is not allowed.

3.3 Performance.

3.3.1 Mobility. The crane shall be capable of being transported in the following positions:

3.3.1.1 Loaded, erected position. The crane shall be capable of lifting a 6,000 pounds load to at least two feet off the ground and capable of being towed for a distance of at least 10 feet over smooth concrete or macadam paving at a speed of not less than two miles per hour.

3.3.1.2 Unloaded, erected position. The crane shall be capable of being towed for a distance of at least three miles over smooth concrete or macadam paving at a speed of not less than ten miles per hour.

3.3.1.3 Unloaded, air transport position. The crane shall be capable of being towed for a distance of at least three miles over smooth concrete or macadam paving at a speed of not less than ten miles per hour.

3.3.2 Capacity.

3.3.2.1 Rated capacity. The crane shall be rated for lifting a 6,000 pound load to a hook height of 17 feet at a reach of 10 feet from the crane mast with the boom horizontal. The crane shall prevent lifting of items greater than the rated load plus 10 percent (6600 lbs).

3.3.2.2 Proof load. The crane shall be capable of supporting a 12,000 pound load to a hook height of 17 feet at a reach of 10 feet from the crane mast with the boom horizontal, with no damage, permanent deformation, or instability.

3.3.3 Operating speed.

3.3.3.1 Lift speed of hook. The lift speed of hook shall not be less than 2.5 feet per minute with a load of 6,000 pounds and shall not exceed 10 feet per minute with no load.

3.3.3.2 Lowering speed of hook. The lowering speed of hook with any load shall not be less than 1.75 feet per minute and shall not exceed 10 feet per minute.

3.3.4 Winch brake performance. The winch brake shall be capable of stopping a 6,000 pound load within a distance of 1 inch while lifting and within a distance of 1 1/2 inches while lowering. The winch brake shall be automatic and shall not require any manual actuation.

3.4. Frame assembly.

3.4.1 Towbar. The frame assembly shall be equipped with a steering towbar designed for towing by vehicle having a pintle hook height of 10 to 30 inches. The towbar shall incorporate a lunette eye (see 6.3.5). The towbar shall be capable of folding onto the crane for storage. A locking device shall be incorporated in the design to lock the towbar in its folding position.

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3.4.2 Casters. The crane shall be equipped with commercial swivel type casters with built in brakes. Each caster shall have a minimum safety factor of three each. Each caster tire shall be polyurethane type. Each caster shall be of sufficient size as to accommodate the requirements of 3.2.10 and to prevent imprint damage to the surface which the crane is operating on

3.4.2.1 Brake system. The built-in brakes of the casters shall be capable of immobilizing the crane in lifting and lowering operation of any load up to 6,000 pounds. The brakes shall be engaged/disengaged by a linked brake system which is hand operated.

3.4.2.2 Swivel locks. All casters on the crane shall be equipped with swivel locks for towing.

3.5 Winch, mast, and boom.

3.5.1 Winch. The winch shall be a 28V direct current (dc) powered unit. Provisions shall be made to allow the power source to be either 28V dc or 120V alternate current (ac) 400 cycle. The winch normal capacity shall be at least 6,000 pounds and the overload capacity of the winch shall be at least 12,000 pounds.

3.5.1.1 Wire rope. The wire rope shall have a strength factor of not less than 3.5 in accordance with J959. Wire rope shall be corrosion resistant and not require lubrication. Sufficient rope shall be provided for all conditions of operation and erection of the crane specified herein.

3.5.1.2 Hook and block. The hook shall have a throat opening of 2 inches minimum and 5 inches maximum. The hook and block shall be of sufficient weight to insure payout of the cable with no load on the hook. The hook shall be of the swivel type, with a minimum factor of safety of four based on ultimate strength. The block assembly shall provide for attachment of crane cable for not less than two part line.

3.5.1.3 Anti-two block system. An anti-two block system shall be provided to prevent cable breakage and wedging or binding of the hook when at boom end.

3.5.2 Mast. The crane mast shall lock securely in the vertical position for operation and land transportation, and shall fold or collapse for storage and air transportation. A self-contained erection system shall be provided.

3.5.3 Boom. The boom shall be capable of being adjusted from 30 to 45 degrees with respect to the horizontal. The adjustment shall be accomplished with the crane erected. It shall require a maximum of 15 minutes to complete.

3.5.4 Outrigger legs. The crane shall be equipped with sufficient outrigger legs to provide required stability and shall conform to the requirement of weight and dimensions (paragraph 3.10). The maximum height of the outrigger shall be 30 inches.

3.6 Finish and protective coatings.

3.6.1 Cleaning, painting, plating, anodic films, and chemical treatments.
Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with manufacturer's standard practices.

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3.6.2 Finish. The paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defects.

3.6.2.1 Forest Green. Unless otherwise specified (see 6.2). the outer surface of the crane and the interior surfaces of compartments shall be painted with standard, commercially available, polyurethane paint in the color commonly known as Forest Green. Forest Green color is used for safety purposes in the continental United States.

3.6.2.2 Desert Sand. When specified (see 6.2), the outer surface of the crane and the interior surfaces of compartments (except in the cab) shall be painted with standard, commercially available, polyurethane paint in a no-gloss tan color. This color is required for safe operation in a desert environment.

3.7 Markings.

3.7.1 Exterior markings. Markings for crane painted Forest Green shall be applied with black paint and markings for cranes painted Desert Sand shall be applied with white paint. Unless otherwise specified (see 6.2), the crane shall be marked for the appropriate service (see 6.3.6).

3.8 Identification plates and operating instructions.

3.8.1 Marking plates. All identification plates, instruction plates, data plates, etc., shall be of a size and shape consistent with the information required thereon.

3.8.2 Identification plate. A permanently marked identification plate shall be securely attached to the crane in a readily accessible location. The material of the plate shall be corrosion resistant. The identification shall contain the following information:

NOMENCLATURE
MANUFACTURER'S MODEL NUMBER
MANUFACTURER'S SERIAL NUMBER
MANUFACTURER'S NAME
MANUFACTURER'S ADDRESS
MANUFACTURER'S SERVICE PHONE NUMBER
DATE OF MANUFACTURE
CONTRACT NUMBER
NATIONAL STOCK NUMBER

3.8.3 Control markings. All controls, valves, gauges, and indicators used in the operation of the crane shall be identified by nameplates securely installed in clearly visible locations.

3.8.4 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed on the unit in a visible location. Instructions shall be clear, concise, and adequate to enable operation of the crane without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.8.5 Transportation data plate. A transportation data plate shall be provided. The plate shall be made from corrosion resistant material. The plate shall contain at least the following information:

a. Side and rear silhouette views of the crane in transport configuration.

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b. Horizontal and vertical location of the center of gravity of the crane in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width, and height.

f. Tiedown instructions for air transport in Types C-130, C-141, C-17, KC-10, KC-135 and C-5 Aircraft.

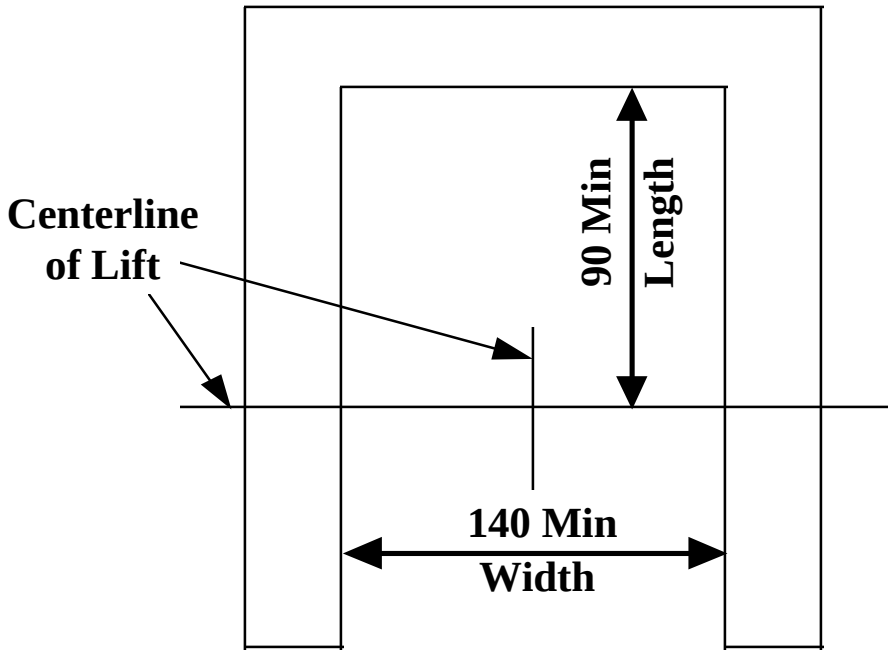
3.9 Part numbering of interchangeable parts. All parts having the same manufacturers part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification (see 6.3.7).

3.10 Weight and dimensions.

3.10.1 Operational weight and dimensions. The crane in fully operational condition, including all equipment and accessories, shall conform to the following weight and dimensions:

a. Over-all length	216 inches (maximum)
b. Over-all width	180 inches (maximum)
c. Over-all height, boom at 45 degree	324 inches (maximum)
boom at 30 degree	276 inches (minimum)
d. Hook height, boom at 45 degree	264 inches (minimum)
boom at 0 degree	204 inches (minimum)
e. Straddle width at centerline of lift with legs spread and boom level	140 inches (minimum)
f. Straddle length at centerline of lift with legs spread and boom level	90 inches (minimum)

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g. Weight (Suitable per 3.2.5) 5000 pounds (maximum).

3.10.2 Transportation weight and dimensions. The crane in the transportation configuration, including all equipment and accessories, shall not exceed the size limitations of two standard aircraft pallets (216.0 by 84.0 inches).

3.11 Fabrication.

3.11.1 Welders and welding.

3.11.1.1 Welders. All welders employed in the fabrication of the crane, shall be certified in accordance with the applicable requirements of the American Welding Society.

3.11.1.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. Welds shall contain sufficient strength as to transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loading.

3.12 Workmanship. The crane, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

4. VERIFICATION

4.1 Classification of tests. The inspection and testing of the crane shall be classified as follows:

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a. First article testing (see 4.3 and 6.2.1).

b. Acceptance testing (see 4.4).

4.2 Test conditions.

4.2.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.2.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

4.2.3 Preparation for tests. The crane, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.2.4 Test rejection criteria. Throughout all tests specified herein, the crane shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including lubricant, or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between crane components or between the crane, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

4.3 First article testing.

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4.3.1 First article test samples. The first article test samples shall be identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced and tested as follows:

a. Contractor test sample (see 4.4.2.1).

b. User test sample (see 4.3.2.2).

4.3.2 First article tests.

4.3.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.5, 4.6, and 4.7.

4.3.2.2 User test sample. The user test sample shall successfully complete the tests in 4.5.1, 4.5.2, and 4.5.9.3 prior to being shipped to the user test facility for the test described in 4.9.

4.3.3 Test report. Upon completion of the first article tests, a first article test report shall be prepared (see 6.3.8). An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles, or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reason being behind any corrections made, to be made, or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used.

g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact.

h. List of special tools and diagnostic equipment necessary to perform field level maintenance.

4.4 Acceptance testing. Each production crane shall be subjected to the tests specified in 4.5.1,.

4.5 Test methods.

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4.5.1 Examination of product. The crane shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting, and markings. For the first article cranes, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.5.2 Proof load test. A load of 12,000 pounds shall be attached to the hook and tested in accordance with 3.3.2.2. The load shall be supported by the crane alone without any external support for a period of not less than 30 minutes. The load shall then be lowered and the crane shall be inspected. Evidence of cracks, permanent distortion, or any other failure shall be cause for rejection.

4.5.3 Load movement test. A load of 6,000 pounds shall be attached to the cable hook and lifted to a height of twenty feet. The crane shall be moved for a distance of 10 feet and then rotated 180 degree by two persons. The force required for movement shall be measured and recorded to demonstrate compliance with 3.2.10.

4.5.4 Operation test. A load of 6,000 pounds shall be attached to the cable hook with the boom fully extended. The load shall be lifted to the maximum hook height and lowered for 10 cycles.

4.5.5 Operating speed. The crane shall be tested during the operation test (4.5.4) to demonstrate compliance with 3.3.3

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4.5.6 Winch brake performance test. The winch brake performance shall be verified during the operation test (4.5.4) to demonstrate compliance with 3.3.4.

4.5.7 Caster brake tests. The caster brakes shall be engaged during the operation test (4.5.4) to demonstrate compliance with 3.4.2.

4.5.8 Mobility tests. Mobility tests shall be performed in accordance with 3.3.1. Any evidence of instability or inability to meet Air Force requirements for towed and manually propelled support equipment (see 6.3.2) shall be cause for rejection.

4.5.9 Design verification.

4.5.9.1 Weight and weight distribution. The weight on each corner of the crane shall be measured with the unit standing on a flat level surface, both in the empty and fully loaded conditions, to demonstrate compliance with 3.2.4 and 3.10.

4.5.9.2 Dimensional verification. The dimensions specified herein shall be measured with the crane standing on a flat level surface to demonstrate compliance with 3.10. Overall height and crane lean shall be measured in both the empty and the fully loaded conditions.

4.5.9.3 Folding/erection verification. Fold/erect the crane three cycles to demonstrate compliance with 3.2.7.

4.5.10 Environmental tests.

4.5.10.1 Low and high temperature storage and operation. The complete crane, including all accessories, shall be tested using an approved test method (see 6.3.9). The crane shall be subjected to the minimum and

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maximum storage and operation temperatures per 3.2.3(a) for a minimum of 24 hours. Operational check-outs shall be in accordance with 4.5.10.4.

4.5.10.2 Rain. The complete crane, including all accessories shall be tested using an approved test method (see 6.3.9) to demonstrate compliance with 3.2.3(c). Operational check-outs shall be in accordance with 4.5.10.4. The interior of all compartments shall be examined for water; any evidence of leakage shall be cause for rejection.

4.5.10.3 Other environmental tests. The components and materials used in the construction of the crane shall be subjected to, or certified as having passed, or being capable of passing, the following tests using an approved test method (see 6.3.9) to demonstrate compliance with 3.2.3(b,d,e). Operational check-outs shall be in accordance with 4.5.10.4.

4.5.10.4 Operational checkouts. All controls, levers, mechanisms, and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation.

4.6 Reliability demonstration. Compliance with the quantitative reliability requirements of 3.2.2.1 shall be demonstrated by the tests in 4.6.1. Scheduled maintenance or adjustments authorized by this specification or the procuring activity shall be not be counted as failures.

4.6.1 Crane durability test. A load of 6,000 pounds shall be attached to the cable hook with the boom fully extended. The load shall be lifted to maximum hook height, stopped and rested for 30 seconds maximum, and lowered to ground level, for 1,610 cycles (1,000 MCBF demonstrated at a 0.8 confidence level).

4.7 Maintainability demonstration. Maintenance task performance shall demonstrate compliance with the maintenance requirements in 3.2.2.2. All scheduled maintenance tasks and all tasks listed in 3.2.2.2.1 shall be performed. Additionally, the procuring activity shall select corrective maintenance tasks from an approved tasks list provided by the contractor in the first article test procedure. No fewer than 25 percent and no more than 50 percent of the total number of tasks on the approved tasks list shall be demonstrated to verify a MTTR within the required limit. As part of this demonstration, use the recommended frequencies of scheduled maintenance tasks and the times recorded to accomplish each task to develop an expected value of scheduled maintenance downtime per operating hour. The capability of performing maintenance tasks by personnel wearing arctic mittens and chemical defense gear shall be demonstrated.

4.8 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.2.5. The evaluation shall also include a dimensional analysis for the crane while loaded aboard Types C-130, C-141, C-17, KC-10, KC-135 and C-5 Aircraft.

4.9 User test. The crane shall be delivered to the location specified (see 6.2) for not less than 30 days of operational use. The test period shall begin after the crane has been inspected and prepared for use and shall be available for up to 240 hours of operations. The crane shall be available for any eight hour period specified by the Government each day, seven days a week. Any period of more than two hours of the same day that the crane is out of service for maintenance or repair shall not be considered an available day of the test period. The crane shall be operated and scheduled maintenance shall be performed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the crane. Any parts or corrective actions required shall be provided by the contractor. Test operations will simulate normal Air Force usage and will not exceed the design capabilities of the crane.

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The sole criteria for acceptance/rejection of the user test shall be based on the performance requirements of this specification.

4.10 Structural Analysis. A structural analysis shall be performed on the load supporting components of the crane assembly to demonstrate the compliance of 3.2.6.

4.11 Refurbishment. After successful completion of all required testing, each crane shall be restored to like new condition.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

6. NOTES

6.1 Intended use. The 6,000 pound crane covered by this purchase description is intended for but not limited to use in removal and installation of Type C-130 aircraft engines and other aircraft components. The crane is intended for use in phase dock hangers, nose dock hangers, and ramp areas. The crane is also intended to be maneuvered by two personnel when unloaded or loaded with a 6,000 pound load.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.2)

c. Finish color required, if not Forest Green or Desert Sand, (see 3.6.2.1 and 3.6.2.2).

d. Markings required, if other than conventional service markings (see 3.7).

e. Location for user testing (see 4.9).

f. Specific mode of packaging, if required (see 5.1)

6.2.1 First article. When a first article inspection is required, the items should be first article samples. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.3 Definitions.

6.3.1 Commercial. The latest model of a contractors standard product having demonstrated industry acceptability by having been manufactured, marketed, and sold in significant quantities to the commercial public for the same or similar application, except as specified herein, for at least 3 years prior to invitation for bids or request for proposal. Proven product improvements incorporated in the commercial production unit during this 3 year period are acceptable. An item is "sold in significant quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes, developmental units, or experimental units, do not meet this requirement. The "commercial public" are buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For purposes of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.3.2 Support equipment requirements. Refer to MIL-STD-1784, Mobility, Towed and Manually Propelled Support Equipment, for detailed information on Air Force requirements for support equipment.

6.3.3 Manual force requirements. Refer to MIL-STD-1472, Human Engineering Design Criteria for Military Systems, Equipment, and Facilities, for detailed information on human interface requirements.

6.3.4 Air Transportability. Refer to MIL-STD-1791, Designing for Internal Aerial Delivery in Fixed Wing Aircraft, for detailed information on interfacing with type C-130, C-141, C-5, and C-17 aircraft.

6.3.5 Towbar. Refer to MS51336, Lunette-coupler, Drawbar, Ring, for detailed information on design of lunette eye.

6.3.6 Exterior Markings. Refer to MIL-STD-1223, Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking, and Data Plate Standards, for detailed information on appropriate service markings.

6.3.7 Part identification. Refer to MIL-STD-130, Identification Marking of US Military Property, for detailed information on marking of component parts.

6.3.8 First Article Report. Refer to MIL-STD-831, Preparation of Test Reports, for detailed information on report format and content.

6.3.9 Environmental Tests. Refer to MIL-STD-810, Environmental Test Methods and Engineering Guidelines, for detailed information on temperature, humidity, moisture, and other environmental tests.

6.3.10 Subject term (key word) listing.

Boom
Casters
Chain
Hook
Hydraulic
Lunette Eye
Mast
Towbar
Wire rope

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Custodians:
Army-AT
Navy-SH
Air Force-99

Preparing Activity:
Air Force-84

Agent: Air Force-99

Project 3950-0318

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-32009	97/07/10

3. DOCUMENT TITLE

CRANE, PORTABLE, FOLDING, 6000 LB CAPACITY, AIRCRAFT ENGINE AND PROPELLER;GENERAL SPECIFICAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
WR-ALC/TILCC	(1) Commercial (912) 926-1183	(2) AUTOVON 468-1183
c. ADDRESS *(Include Zip Code)* 225 OCMULGEE COURT ROBBINS AFB, GA 31098-1647	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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